Signals and Systems Lecture-12: Fourier Transform (3)

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Multiplication-Convolution Duality (1)

The Fourier Transform of the convolution of two CT signals verifies the following relation:

$$x(t) * y(t) \rightleftharpoons X(f)Y(f)$$

- In other words:

Convolution in the time domain

$$\Updownarrow$$

Multiplication in the frequency domain

$$x(t) * y(t) \rightleftharpoons X(f)Y(f)$$
$$x(t)y(t) \rightleftharpoons X(f) * Y(f)$$

- This can significantly simplify the evaluation of convolutions.

Multiplication-Convolution Duality (2)

Example: Find the FT of $x(t) = \text{tri}\,(t)$.

- From lecture 6, the triangular function satisfies the following relation:

$$\text{tri}\,(t) = \text{rect}\,(t) * \text{rect}\,(t)$$

- Taking the FT of both sides of the above equation:

$$\begin{aligned}\mathcal{F}\left[\text{tri}\,(t)\right] &= \mathcal{F}\left[\text{rect}\,(t)\right]\mathcal{F}\left[\text{rect}\,(t)\right] \\ &= \text{sinc}(f).\text{sinc}(f) \\ &= \text{sinc}^2(f)\end{aligned}$$

  - Note that $\text{tri}\,(t)$ is real and even while $\text{sinc}^2(f)$ is real and even.

Multiplication-Convolution Duality (3)

Example: Find the FT of the AM signal

$$y(t) = x(t)\cos(2\pi f_c t)$$

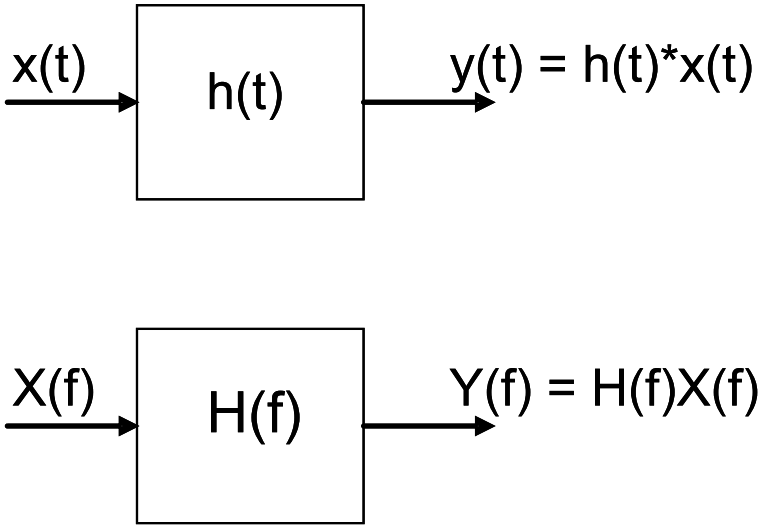
where f_c is the carrier frequency

- Taking the FT of both sides of the above equation:

$$\begin{aligned} Y(f) &= X(f) * \mathcal{F}\left[\cos(2\pi f_c t)\right] \\ &= X(f) * \left[\frac{1}{2}\delta(f - f_c) + \frac{1}{2}\delta(f + f_c)\right] \\ &= \frac{1}{2}X(f - f_c) + \frac{1}{2}X(f + f_c) \end{aligned}$$

  - which is the same as the expression obtained before.

Frequency Response of LTI systems (1)

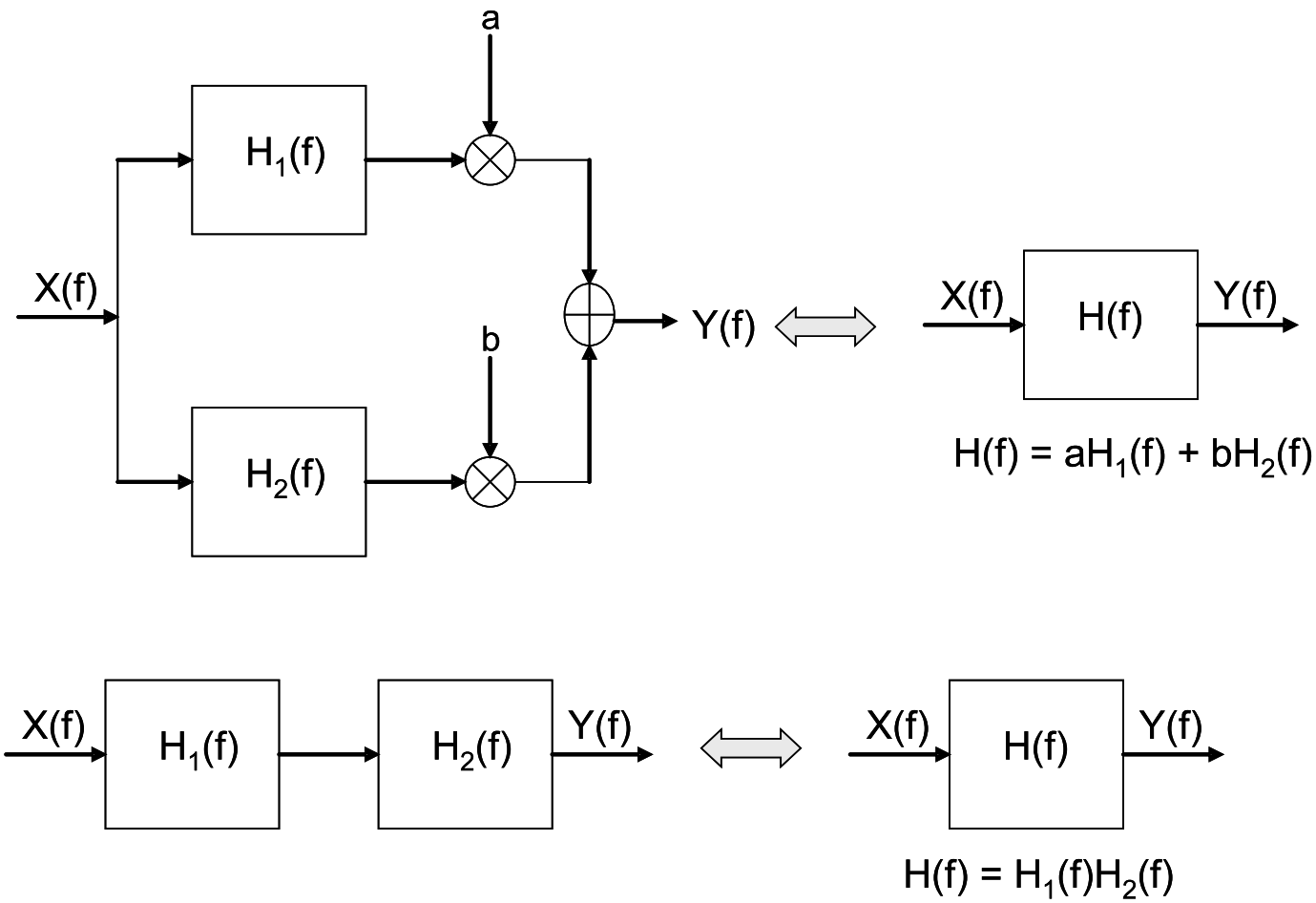


- $h(t)$ is the impulse response of the LTI system.
- $H(f)$ is the frequency response of the LTI system.
- $H(f)$ is the Fourier transform of $h(t)$:

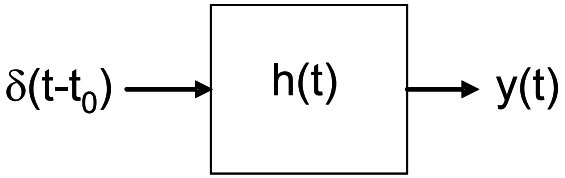
$$h(t) \rightleftharpoons H(f) = \mathcal{F}[h(t)]$$

Frequency Response of LTI systems (2)

- LTI systems can be studied in the frequency domain.



Example 1



- Time Domain:

$$y(t) = h(t) * \delta(t - t_0) = h(t - t_0)$$

- Frequency Domain:

$$\begin{aligned} h(t) &\rightleftharpoons H(f) \\ \delta(t - t_0) &\rightleftharpoons e^{-j2\pi t_0 f} \\ y(t) &\rightleftharpoons Y(f) = H(f)e^{-j2\pi t_0 f} \end{aligned}$$

- Thus the time shifting property is rederived:

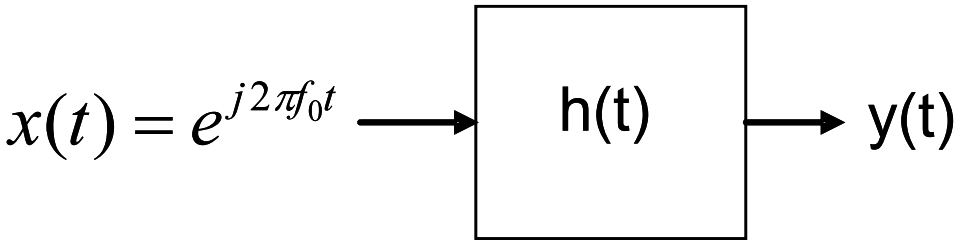
$$h(t - t_0) \rightleftharpoons H(f)e^{-j2\pi t_0 f}$$

Example 2

The frequency shifting property (revisited):

$$
\begin{aligned}
y(t) &= x(t)e^{j2\pi f_0 t} \\
Y(f) &= X(f) * \mathcal{F}\left[e^{j2\pi f_0 t}\right] \\
&= X(f) * \delta(f - f_0) \\
&= X(f - f_0)
\end{aligned}
$$

Example 3



- Following from the frequency shifting property of the Fourier Transform:

$$\begin{aligned} Y(f) &= H(f)X(f) \\ &= H(f)\delta(f - f_0) \\ &= H(f_0)\delta(f - f_0) \end{aligned}$$

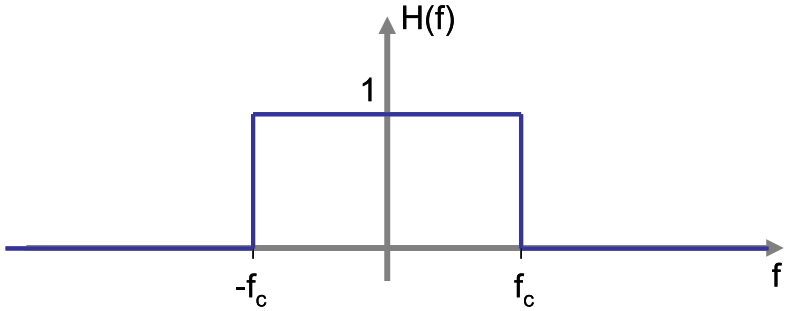
- Consequently, by applying the inverse Fourier Transform:

$$\begin{aligned} Y(f) &= H(f_0)\delta(f - f_0) \\ \Rightarrow \quad y(t) &= H(f_0)e^{j2\pi f_0 t} \end{aligned}$$

- The gain introduced by a LTI system at the frequency component f_0 is equal to $H(f_0)$.

Impulse Response of an Ideal Lowpass Filter (1)

Impulse Response of an Ideal Lowpass Filter:



- *Method 1:*

$$h(t) = \int_{-\infty}^{+\infty} H(f) e^{j2\pi ft} \mathrm{d}f = \int_{-f_c}^{+f_c} e^{j2\pi ft} \mathrm{d}f$$

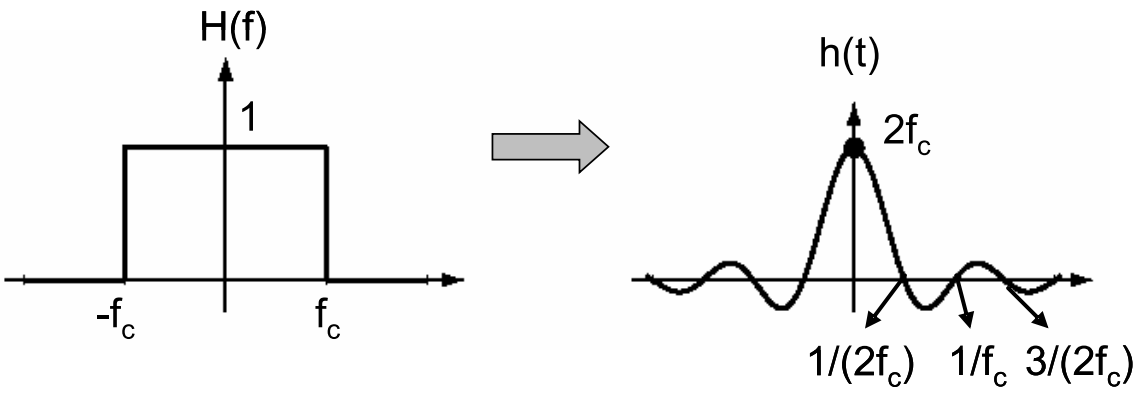
$$= \frac{1}{j2\pi t} \left[e^{j2\pi ft} \right]_{-f_c}^{+f_c} = \frac{1}{\pi t} \sin\left(2\pi f_c t\right) = 2f_c \mathrm{sinc}\left(2f_c t\right)$$

- *Method 2:*

$$\mathrm{sinc}(t) \rightleftharpoons \mathrm{rect}(f)$$

$$\Rightarrow \quad 2f_c \mathrm{sinc}(2f_c t) \rightleftharpoons \mathrm{rect}\left(\frac{f}{2fc}\right)$$

Impulse Response of an Ideal Lowpass Filter (2)



- $h(0)$ must be equal to $\int_{-\infty}^{+\infty} H(f)\mathrm{d}f$.
  - Since $h(t) = 2f_c\mathrm{sinc}(2f_c t)$, then for $t = 0$: $h(0) = 2f_c\mathrm{sinc}(0) = 2f_c$.
  - Since $H(f) = \mathrm{rect}\left(\frac{f}{2f_c}\right)$, then $\int_{-\infty}^{+\infty} H(f)\mathrm{d}f$ is the surface under $H(f)$ which is equal to $2f_c$.
  - Consequently, the equality is verified.
- Note that $h(t)$ is not equal to zero for negative values of t and the system is not causal.

Response of real LTI systems to sinusoids (1)

Proposition: The response of a real LTI system with transfer function $H(f)$ to $\cos(2\pi f_0 t)$ is given by:

$$y(t) = |H(f_0)| \cos(2\pi f_0 t + \arg[H(f_0)])$$

Since the LTI system is real:

- The impulse response $h(t)$ is real.
- The transfer function $H(f)$ verifies: $H(-f) = H^*(f)$.

Response of real LTI systems to sinusoids (2)

Proof:

$$x_1(t) = e^{j2\pi f_0 t} \rightarrow y_1(t) = H(f_0)e^{j2\pi f_0 t}$$
$$x_2(t) = e^{-j2\pi f_0 t} \rightarrow y_2(t) = H(-f_0)e^{-j2\pi f_0 t} = H^*(f_0)e^{-j2\pi f_0 t}$$

- Since $\cos(2\pi f_0 t) = \frac{1}{2}e^{j2\pi f_0 t} + \frac{1}{2}e^{-j2\pi f_0 t}$:

$$\begin{aligned}
\cos(2\pi f_0 t) &\rightarrow \frac{1}{2}\left[H(f_0)e^{j2\pi f_0 t} + H^*(f_0)e^{-j2\pi f_0 t}\right] \\
&= \frac{1}{2}|H(f_0)|\left[e^{j\arg[H(f_0)]}e^{j2\pi f_0 t} + e^{j\arg[H^*(f_0)]}e^{-j2\pi f_0 t}\right] \\
&= \frac{1}{2}|H(f_0)|\left[e^{j\arg[H(f_0)]}e^{j2\pi f_0 t} + e^{-j\arg[H(f_0)]}e^{-j2\pi f_0 t}\right] \\
&= \frac{1}{2}|H(f_0)|\left[e^{j[\arg[H(f_0)]+2\pi f_0 t]} + e^{-j[\arg[H(f_0)]+2\pi f_0 t]}\right] \\
&= |H(f_0)|\cos(2\pi f_0 t + \arg[H(f_0)])
\end{aligned}$$

Time Differentiation (1)

Proposition: The FT of the time derivative of $x(t)$ is given by:

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} \rightleftharpoons (j2\pi f)X(f)$$

- **Proof**: Applying the inverse FT, $x(t)$ can be written as:

$$x(t) = \int_{-\infty}^{+\infty} X(f)e^{j2\pi ft}\mathrm{d}f$$

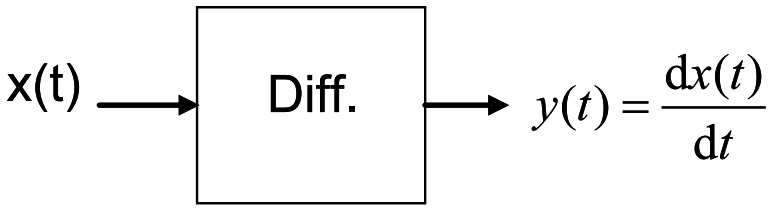
- Differentiating both sides w.r.t. time:

$$\begin{aligned}\frac{\mathrm{d}x(t)}{\mathrm{d}t} &= \int_{-\infty}^{+\infty} (j2\pi f)X(f)e^{j2\pi ft}\mathrm{d}f \\ &= \mathcal{F}^{-1}[(j2\pi f)X(f)]\end{aligned}$$

- In the same way, differentiating n times:

$$\frac{\mathrm{d}^n x(t)}{\mathrm{d}t^n} \rightleftharpoons (j2\pi f)^n X(f)$$

Time Differentiation (2)

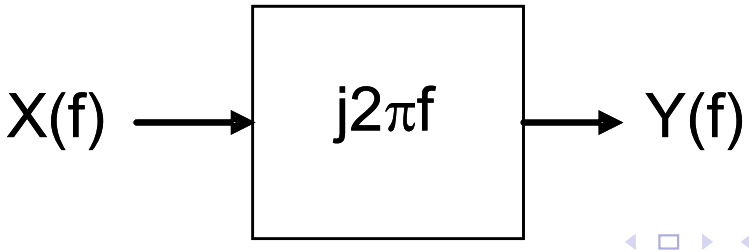


- The transfer function of a differentiator can be calculated from:

$$y(t) = \frac{dx(t)}{dt}$$
$$\Rightarrow \quad Y(f) = (j2\pi f)\,X(f)$$

- Therefore, the transfer function (or frequency response) of a differentiator is:

$$H(f) = j2\pi f$$



Time Differentiation (3)

- Consequently:

$$|Y(f)| = |2\pi f|\,|X(f)|$$
$$\arg[Y(f)] = \arg[X(f)] \pm \frac{\pi}{2}$$

- A differentiator enhances sharp edges since the gain is higher for higher frequencies.
- It introduces a $\frac{\pi}{2}$ (resp. $-\frac{\pi}{2}$) phase shift for positive (resp. negative) frequencies.

Example 4 (1)

- $X(f)$:

$$x(t) = \text{rect}(t) \quad \rightleftharpoons \quad X(f) = \text{sinc}(f)$$

- $Y(f)$:

$$Y(f) = (j2\pi f)X(f) = (j2\pi f)\text{sinc}(f) = (j2\pi f)\frac{\sin(\pi f)}{\pi f} = 2j\sin(\pi f)$$

$$= 2j\left[\frac{e^{j\pi f} - e^{-j\pi f}}{2j}\right]$$

$$= e^{j\pi f} - e^{-j\pi f}$$

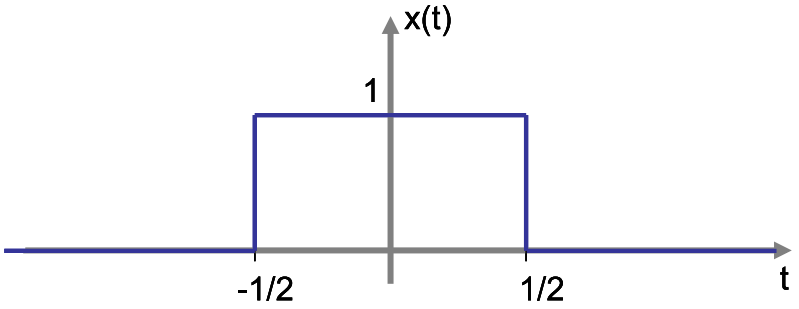
Example 4 (2)

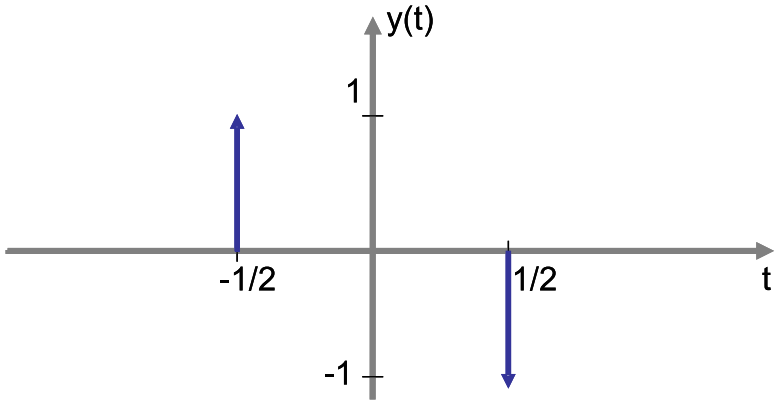
From the time shifting property $x(t-t_0) \rightleftharpoons X(f)e^{-j2\pi t_0 f}$:

$$Y(f) = e^{-j2\pi(-\frac{1}{2})f} - e^{-j2\pi(\frac{1}{2})f}$$

$$\downarrow\uparrow$$

$$y(t) = \delta(t+\frac{1}{2}) - \delta(t-\frac{1}{2})$$





Example 4 (3)

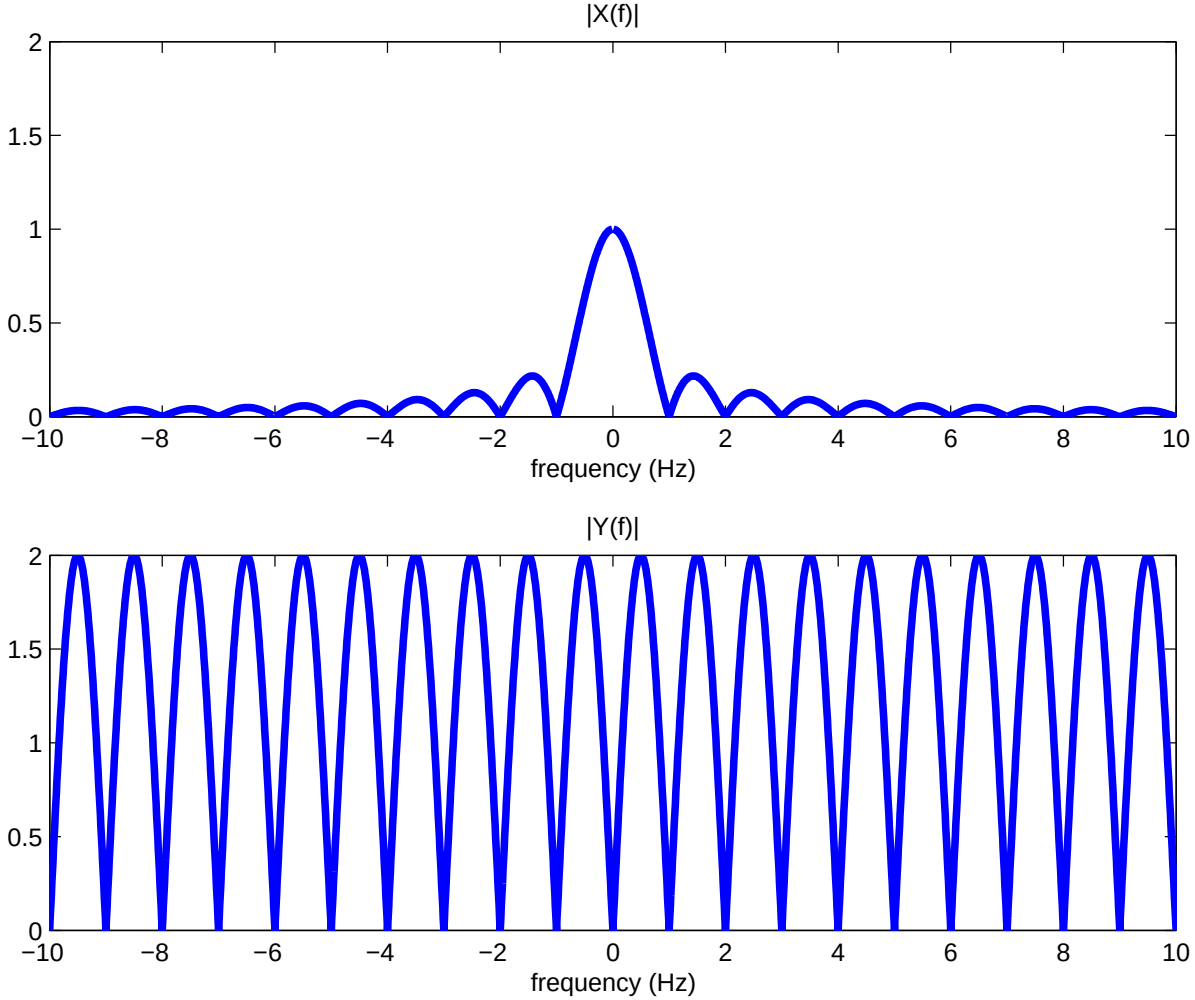
- $y(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t}$ and its Fourier transform are given by:

$$y(t) = \delta(t + \frac{1}{2}) - \delta(t - \frac{1}{2})$$
$$\Updownarrow$$
$$Y(f) = 2j\sin(\pi f)$$

  - Note that $y(t)$ is real and odd while $Y(f)$ is pure imaginary and odd.
  - On the other hand, $Y(0) = 2j\sin(2\pi 0) = 0$ and:

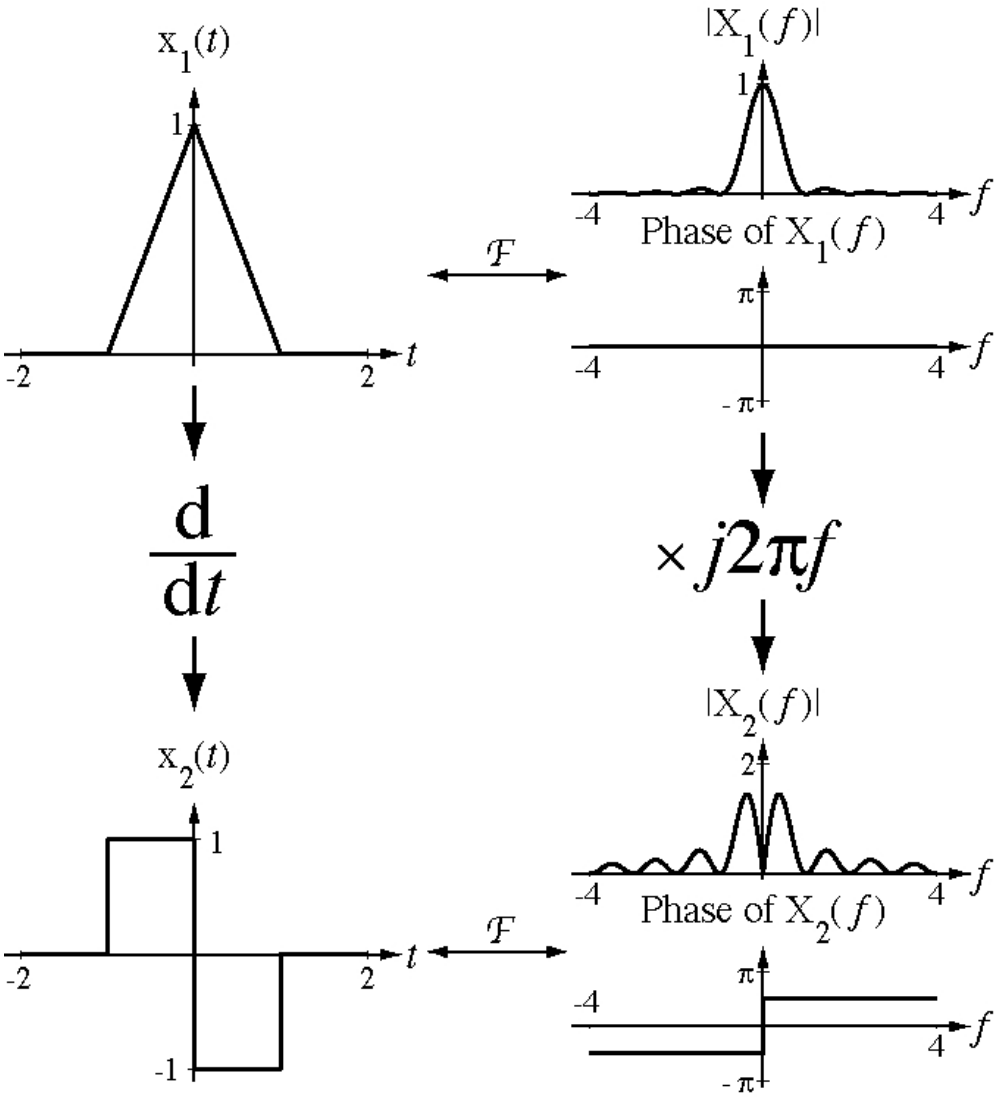
$$\int_{-\infty}^{+\infty} y(t)\mathrm{d}t = \underbrace{\int_{-\infty}^{+\infty} \delta(t + \frac{1}{2})\mathrm{d}t}_{=1} - \underbrace{\int_{-\infty}^{+\infty} \delta(t - \frac{1}{2})\mathrm{d}t}_{=1} = 0 = Y(0)$$

Example 4 (4)



- Note that the high frequency components are enhanced.

Example 5 (1)


x1(t)
1
-2
2
t
|X1(f)|
1
-4
4
f
Phase of X1(f)
π
-π
-4
4
f
F
d/dt
× j2πf
x2(t)
1
-1
-2
2
t
|X2(f)|
2
-4
4
f
Phase of X2(f)
π
-π
-4
4
f
F

Example 5 (2)

- In this example:

$$x_1(t) = \text{tri}(t) \;\rightleftharpoons\; X_1(f) = \text{sinc}^2(f)$$

  - $x_1(t)$ is real and even and $X_1(f)$ is also real and even.
  - $\arg[X_1(f)] = 0$ for all values of f.
- To verify the time-differentiation property of the FT, the FT of $x_2(t) = \frac{\mathrm{d}x_1(t)}{\mathrm{d}t}$ will be calculated by two different methods.
- By applying the time-differentiation property:

$$\begin{aligned} X_2(f) &= (j2\pi f)X_1(f) \\ &= (j2\pi f)\text{sinc}^2(f) \\ &= (j2\pi f)\text{sinc}(f)\frac{\sin(\pi f)}{\pi f} \\ &= 2j\text{sinc}(f)\sin(\pi f) \end{aligned}$$

  - $x_2(t)$ is real and odd and $X_2(f)$ is pure imaginary and odd.
  - Since $x_2(t)$ is real, then $|X_2(f)| = 2|\text{sinc}(f)||\sin(\pi f)|$ is even.

Example 5 (3)

- By direct calculations:
  - for $t \leq -1$, $x_1(t) = 0$ and $x_2(t) = \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = 0$.
  - for $-1 \leq t \leq 0$, $x_1(t) = t + 1$ and $x_2(t) = \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = 1$.
  - for $0 \leq t \leq 1$, $x_1(t) = -t + 1$ and $x_2(t) = \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = -1$.
  - for $t \geq 1$, $x_1(t) = 0$ and $x_2(t) = \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = 0$.
- Consequently:

$$
\begin{aligned}
x_2(t) &= \mathrm{rect}\left(t + \frac{1}{2}\right) - \mathrm{rect}\left(t - \frac{1}{2}\right) \\
\Rightarrow \quad X_2(f) &= \mathrm{sinc}(f)e^{j2\pi f\frac{1}{2}} - \mathrm{sinc}(f)e^{-j2\pi f\frac{1}{2}} \\
&= \mathrm{sinc}(f)\left[e^{j\pi f} - e^{-j\pi f}\right] \\
&= 2j\mathrm{sinc}(f)\left[\frac{e^{j\pi f} - e^{-j\pi f}}{2j}\right] \\
&= 2j\mathrm{sinc}(f)\sin(\pi f)
\end{aligned}
$$

  - Note that the same result is obtained.

Example 5 (4)

- Note that:

$$X_2(f) = (j2\pi f)X_1(f)$$

- For $f \geq 0$:

$$\begin{aligned}\arg[X_2(f)] &= \arg[X_1(f)] + \arg[j2\pi f] \\ &= 0 + \frac{\pi}{2} = \frac{\pi}{2}\end{aligned}$$

- For $f \leq 0$:

$$\begin{aligned}\arg[X_2(f)] &= \arg[X_1(f)] + \arg[j2\pi f] \\ &= 0 - \frac{\pi}{2} = -\frac{\pi}{2}\end{aligned}$$

- Finally:

$$\int_{-\infty}^{+\infty} x_1(t)\mathrm{d}t = 1 = X_1(0)$$

$$\int_{-\infty}^{+\infty} x_2(t)\mathrm{d}t = 0 = X_2(0)$$

Example 6 (1)

Evaluate $y(t) = x(t) * h(t)$ when $x(t) = 2\frac{\sin(4\pi t)}{\pi t}$ and $h(t) = \frac{\sin(8\pi t)}{3\pi t}$.

- *Step 1:* Determine $X(f)$:

$$x(t) = 2\frac{\sin(4\pi t)}{\pi t} = 8\frac{\sin(4\pi t)}{4\pi t} = 8\text{sinc}(4t)$$

Consequently:

$$\text{sinc}(t) \rightleftharpoons \text{rect}(f)$$

$$\Rightarrow \quad x(t) = 8\text{sinc}(4t) \rightleftharpoons 8\frac{1}{4}\text{rect}\left(\frac{f}{4}\right) = 2\text{rect}\left(\frac{f}{4}\right)$$

- *Step 2:* Determine $H(f)$:

$$h(t) = \frac{\sin(8\pi t)}{3\pi t} = \frac{8}{3}\frac{\sin(8\pi t)}{8\pi t} = \frac{8}{3}\text{sinc}(8t)$$

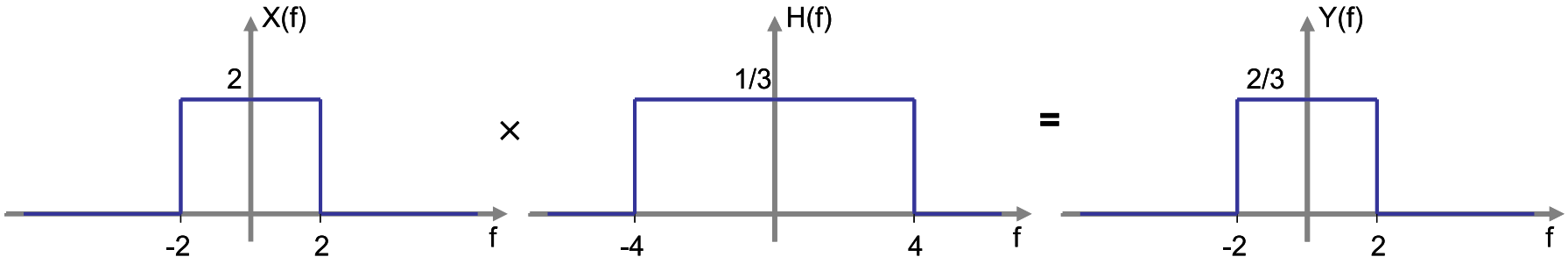
Consequently:

$$h(t) = \frac{8}{3}\text{sinc}(8t) \rightleftharpoons \frac{8}{3}\frac{1}{8}\text{rect}\left(\frac{f}{8}\right) = \frac{1}{3}\text{rect}\left(\frac{f}{8}\right)$$

Example 6 (2)

$$y(t) = x(t) * h(t) \;\Leftrightarrow\; Y(f) = X(f)H(f)$$

$$y(t) = 2\frac{\sin(4\pi t)}{\pi t} * \frac{\sin(8\pi t)}{3\pi t} \;\Leftrightarrow\; Y(f) = 2\text{rect}\left(\frac{f}{4}\right).\frac{1}{3}\text{rect}\left(\frac{f}{8}\right)$$



- Consequently:

$$Y(f) = \frac{2}{3}\text{rect}\left(\frac{f}{4}\right)$$

$$\Leftrightarrow\; y(t) = \mathcal{F}^{-1}[Y(f)] = \frac{2}{3}\left(4\text{sinc}(4t)\right) = \frac{8}{3}\text{sinc}(4t)$$

Integration (1)

The integration property of the FT is given by:

$$\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \;\rightleftharpoons\; \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\delta(f)$$

Example: calculate the FT of the unit step function $u(t)$.

- $u(t)$ can be written as:

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\mathrm{d}\tau$$

- Consequently:

$$\begin{aligned} U(f) &= \frac{\mathcal{F}[\delta(t)]}{j2\pi f} + \frac{1}{2}\mathcal{F}[\delta(t)]_{f=0}\delta(f) \\ &= \frac{1}{j2\pi f} + \frac{1}{2}\delta(f) \end{aligned}$$

  - Note that the second equality follows since $\mathcal{F}[\delta(t)] = 1$.

Integration (2)

Example: calculate the FT of the signum function:

$$\mathrm{sgn}(t) = \begin{cases} 1, & t \geq 0; \\ -1, & t < 0. \end{cases}$$

- $\mathrm{sgn}(t)$ can be written as:

$$\mathrm{sgn}(t) = 2u(t) - 1$$

- Consequently:

$$\begin{aligned} \mathcal{F}[\mathrm{sgn}(t)] &= 2\mathcal{F}[u(t)] - \delta(f) \\ &= 2\left[\frac{1}{j2\pi f} + \frac{1}{2}\delta(f)\right] - \delta(f) = \frac{1}{j\pi f} \end{aligned}$$

- Note that by duality:

$$\begin{aligned} \frac{1}{j\pi t} &\rightleftharpoons \mathrm{sgn}(-f) = -\mathrm{sgn}(f) \\ \Rightarrow \quad \frac{1}{\pi t} &\rightleftharpoons -j\mathrm{sgn}(f) \end{aligned}$$